Endpoint square functions of general Ritt operators on L^1 need not be weak type $(1,1)$

Candidate full counterexample to Open Question 1 of arXiv:2507.07256

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Status. Candidate full counterexample, likely valid. One fixed Ritt operator on a probability-space L^1 makes the endpoint generalized square function fail weak type $(1,1)$ for every $1 \le s < 2$ with $r = 1$ and $\alpha = s - 1$. Thus the universal endpoint question has a negative answer.

1 The source question

For a Ritt operator T on a Banach function space, Hults and Reinhold-Larsson define

$$\mathbf{Q}^T_{\alpha,s,r} f = \left(\sum_{n=1}^{\infty} n^{\alpha} |T^n (I - T)^r f|^s \right)^{1/s}.$$

Their Theorem 1.7 proves strong boundedness when $sr > \alpha + 1$. Open Question 1 asks whether $\mathbf{Q}^T_{\alpha,s,r}$ is of weak type $(1,1)$ on the endpoint $sr = \alpha + 1$ [1, p. 3]. The exact question and its context are reproduced in Figure 1.

2 Proof intuition

A Ritt estimate controls each time increment in operator norm, but it does not force weak-$(1,1)$ control after absolute values are accumulated over many widely separated time scales. We assign the mth dyadic martingale difference the eigenvalue $1 - 10^{-6m}$. Its contribution to the square function is concentrated near time 10^{6m}, so disjoint time blocks recover one uniformly positive contribution from each martingale level. The extreme lacunarity makes all other levels negligible in that block.

For the Rademacher polynomial $f_N = r_1 + \cdots + r_N$, this produces a pointwise lower bound of order $N^{1/s}$. Yet $\|f_N\|_1 \le \sqrt{N}$. The two rates contradict weak type $(1,1)$ whenever $s < 2$. The same lacunarity also makes the operator easy to audit: bounded-variation formulas for martingale multipliers prove both power boundedness and the Ritt difference estimate.

3 Construction and operator estimates

Let

$$\Omega = \{-1, 1\}^{\mathbb{N}}$$

with product probability measure. Write $r_m(\omega) = \omega_m$, let $\mathcal{F}_m = \sigma(r_1, \dots, r_m)$, and denote conditional expectation by $E_m = \mathbb{E}(\,\cdot \mid \mathcal{F}_m)$, with $E_0 = \mathbb{E}$. Put $D_m = E_m - E_{m-1}$.

We first record the elementary multiplier estimate used throughout.

theorem focused on $0 < r < 1$ but their methods apply to any $r > 0$. They also showed item d) for $n_k = 2^k$ but their proof applies to sequences with $n_{k+1} - n_k \sim n_k$ with minor modifications.

In martingale theory, the equivalence between the square function and associate maximal functions is a classical result. This naturally raises the question of whether such an equivalence holds in other contexts. Cohen, Cuny and Lin [6] established that a similar equivalence holds in the context of Ritt operators.

Theorem 1.6. *Let $(X, \mathcal{B}, m)$ be a σ-finite measure space, $1 < p < \infty$ and T a positive contraction on $L^p(X, m)$. Then the following are equivalent:*

a) $\sup_n n\|T^n - T^{n+1}\| < \infty$,
b) *there exists a constant $C_p > 0$ such that $\|\mathbf{Q_{1,2,1}}f\|_p \le C_p\|f\|_p$.*

Let $X = (X, \beta, m)$ be a σ–finite measure space. Throughout these notes, Y denotes a Banach function space, that is, a Banach space of measurable complex–valued functions on X. In particular, this class includes $L^1(X)$ and Orlicz spaces.

Theorem 1.7. *Let T be a Ritt operator on a Banach function space Y. Let $s \ge 1$ and $r > 0$, the following hold:*

a) *If $sr > \alpha + 1$, then $\mathbf{Q^T_{\alpha,s,r}}f$ is bounded on Y;*
b) *If $\beta < r$ and $\{n_k\} \subset \mathbb{N}$ an increasing sequence with $n_{k+1} - n_k \sim n_k$, then $\sup_n n^\beta|T^n(I - T)^r f|$ and $\left(\sum_k \sup_{n_k<n<n_{k+1}} n^{\beta s}|T^n(I - T)^r f|^s\right)^{1/s}$ are bounded on Y, $\lim_{n\to\infty} n^r\|T^n(I - T)^r f\| = 0$ and $\lim_{n\to\infty} n^\beta|T^n(I - T)^r f| = 0$ a.e.;*
c) *If $\beta < r$ and $s \ge 1$ then both $\|n^\beta T^n(I - T)^r f\|_{v(s)}$ and $\|n^\beta T^n(I - T)^r f\|_{o(s)}$ are bounded on Y;*
d) *Let $\{n_k\}$ be any increasing sequence with $n_{k+1} - n_k \sim n_k^\gamma$, for some $\gamma \in (0, 1]$. If $\beta < r + (1 - \gamma)(1 - 1/s)$ then $\left(\sum_k n_k^{\beta s} \max_{n_k\le n,m\le n_{k+1}} |(T^n - T^m)(I - T)^r f|^s\right)^{1/s}$ and $\left(\sum_k n_k^{\beta s}|(T^{n_k} - T^{n_{k+1}})(I - T)^r f|^s\right)^{1/s}$, are bounded on Y. In particular, these results holds for $\beta < r$ and $s \ge 1$ or $\beta \le r > 0$ and $s > 1 > \gamma$.*

Remark. By Theorem 1.6, $\mathbf{Q_{1,2,1}}f$ is bounded on L^p for any $p > 1$, but on L^1, $\mathbf{Q_{1,s,1}}f$ is bounded for $s > 2$, and in general, $\mathbf{Q_{r,s,r}}f$ is bounded on L^1 for $s > 1 + 1/r$. More generally, by Theorem 1.5, $\mathbf{Q_{2r-1,2,r}}f$ is bounded on L^p for any $p > 1$ but on L^1, $\mathbf{Q_{2r-1,s,r}}f$ bounded for any $s > 2$, and $\mathbf{Q_{\alpha,2,r}}f$ is bounded for any $\alpha < 2r - 1$.

Open Question 1. Is $\mathbf{Q_{\alpha,s,r}}f$ of weak type (1,1) for $sr = \alpha + 1$?

Open Question 2. For what range of s are $\|T^n f\|_{v(s)}$ and $\|T^n f\|_{o(s)}$ of weak type (1,1)? And in general, for what range of s are $\|n^r T^n(I - T)^r f\|_{v(s)}$ and $\|n^r T^n(I - T)^r f\|_{o(s)}$ of weak type (1,1)?

Estimates in L^1 present more challenges. For the maximal function $Mf = \sup_{n\ge1} |T^n f|$ only partial results are available, showing that Mf is a weak (1,1) operator for a restricted class of operators acting on probability spaces [3, 27, 7]. In what follows (and in Section 5), we provide answers for the above questions under assumptions analogous to those imposed in the study of Mf.

For the rest of this section $X = (X, \mathcal{B}, m)$ denotes a probability space and τ an invertible measure–preserving transformation on X. T is the composition operator $Tf = f \circ \tau$. To

Figure 1: Full-width rendering of PDF page 3 of arXiv:2507.07256v2. Open Question 1 appears below Theorem 1.7 and its following remark.

Lemma 1 (bounded-variation martingale multipliers). *If $a_m \to 0$ and $\sum_m |a_m - a_{m+1}| < \infty$, then*

$$M_a = \sum_{m=1}^{\infty} a_m D_m$$

converges in operator norm on $L^1(\Omega)$ and

$$\|M_a\|_{1\to 1} \le |a_1| + \sum_{m=1}^{\infty} |a_m - a_{m+1}|. \tag{1}$$

Proof. For a finite truncation, summation by parts gives

$$\sum_{m=1}^{M} a_m D_m = a_M E_M + \sum_{m=1}^{M-1} (a_m - a_{m+1}) E_m - a_1 E_0.$$

Every E_m is a contraction on L^1. The displayed formula is Cauchy in operator norm and yields (1). □

Fix

$$q = 10^{-6}, \qquad \delta_m = q^m, \qquad \lambda_m = 1 - \delta_m,$$

and define

$$T = I - \sum_{m=1}^{\infty} \delta_m D_m = E_0 + \sum_{m=1}^{\infty} \lambda_m D_m. \tag{2}$$

The first series is norm convergent by Lemma 1. The martingale decomposition shows that T^n has eigenvalue 1 on constants and λ_m^n on $D_m L^1$.

Lemma 2. *The operator T is Ritt on $L^1(\Omega)$; more precisely,*

$$\sup_{n\ge 1} \|T^n\|_{1\to 1} \le 3, \qquad \sup_{n\ge 1} n \left\|T^n - T^{n+1}\right\|_{1\to 1} < \frac{2}{e}.$$

Proof. The coefficients $c_m = 1 - \lambda_m^n$ decrease to zero. Lemma 1 therefore gives $\|I - T^n\| \le 2c_1 \le 2$, proving the first estimate.

The multiplier coefficients of $nT^n(I-T)$ are

$$b_m = n\delta_m (1-\delta_m)^n.$$

The scalar function $x \mapsto nx(1-x)^n$ has a unique maximum at $x = 1/(n+1)$. Since (δ_m) decreases, $(b_m)_m$ is unimodal and tends to zero. Lemma 1 and total variation give

$$n\left\|T^n(I-T)\right\| \le 2 \max_m b_m \le 2 \max_{0\le x\le 1} nx(1-x)^n = 2\left(\frac{n}{n+1}\right)^{n+1} < \frac{2}{e}.$$

□

4 Endpoint failure

Theorem 3. *For the single Ritt operator T in* (2) *and every $1 \le s < 2$, the endpoint square function*

$$\mathbf{Q}^T_{s-1,s,1} f = \left(\sum_{n=1}^{\infty} n^{s-1} |T^n(I-T)f|^s \right)^{1/s}$$

is not of weak type $(1,1)$*. In particular, Open Question 1 has a negative answer for general Ritt operators on L^1.*

Proof. Let $L_m = \delta_m^{-1} = 10^{6m}$ and introduce the disjoint integer blocks

$$B_m = \{L_m, L_m + 1, \dots, 2L_m - 1\}.$$

For $N \ge 1$, put $f_N = \sum_{j=1}^N r_j$. Since $D_j f_N = r_j$,

$$T^n(I-T)f_N = \sum_{j=1}^{N} \delta_j \lambda_j^n r_j. \tag{3}$$

Fix $m \le N$. The weighted $\ell^s(B_m)$ norm of the mth summand is at least $1/16$. Indeed, $n \ge L_m$, and

$$\lambda_m^{L_m} = (1 - L_m^{-1})^{L_m} > \frac{1}{4}$$

because $L_m \ge 2$. Hence $\lambda_m^n \ge \lambda_m^{2L_m} \ge 1/16$ on B_m, and

$$\left(\sum_{n \in B_m} n^{s-1} |\delta_m \lambda_m^n|^s \right)^{1/s} \ge \frac{1}{16}. \tag{4}$$

We bound interference in the same norm. If $j > m$, then $\lambda_j^n \le 1$ and

$$\left(\sum_{n \in B_m} n^{s-1} |\delta_j \lambda_j^n|^s \right)^{1/s} \le 2L_m \delta_j = 2q^{j-m}. \tag{5}$$

If $j < m$, set $R = \delta_j/\delta_m = q^{-(m-j)}$. Since $\lambda_j^n \le e^{-n\delta_j} \le e^{-R}$ on B_m, the same calculation gives

$$\left(\sum_{n \in B_m} n^{s-1} |\delta_j \lambda_j^n|^s \right)^{1/s} \le 2Re^{-R}. \tag{6}$$

The bounds are uniform for $1 \le s < 2$. Moreover,

$$2 \sum_{k \ge 1} q^k + 2 \sum_{k \ge 1} q^{-k} e^{-q^{-k}} < 10^{-5} < \frac{1}{32}.$$

For the second sum one may use $q^{-k} \ge k/q$ and monotonicity of xe^{-x} for $x \ge 1$.

The reverse triangle inequality in weighted $\ell^s(B_m)$, applied to (3)–(6), now gives a lower bound $1/32$ for every block and every $\omega \in \Omega$. Since the blocks are disjoint,

$$\mathbf{Q}^T_{s-1,s,1} f_N(\omega) \ge \frac{N^{1/s}}{32} \qquad (\omega \in \Omega). \tag{7}$$

On the other hand, Rademacher orthogonality and Cauchy–Schwarz give

$$\|f_N\|_1 \le \|f_N\|_2 = \sqrt{N}. \tag{8}$$

If the square function were weak type $(1,1)$ with constant C, applying the distributional estimate at $t = N^{1/s}/64$ would yield

$$1 \le \frac{C\,\|f_N\|_1}{t} \le 64CN^{1/2-1/s}.$$

This is impossible as $N \to \infty$ because $s < 2$. □

5 Scope, verification, and novelty

The same fixed operator settles an interval of endpoint parameters, not only one triple: $r = 1$, $1 \le s < 2$, and $\alpha = s - 1$ always satisfy $sr = \alpha + 1$. Thus it refutes the universal question as written. The argument becomes scale-balanced at $s = 2$, and T is not asserted to be a positive contraction or a convolution operator. Accordingly, this packet does not settle $s = 2$ or the structured subclasses for which the source gives positive results.

Eight focused audits separately checked norm convergence of (2), power boundedness, the Ritt estimate, target-block mass, past and future interference, the Rademacher weak-type contradiction, parameter scope, and novelty. The included checker validates the scalar constants and sampled multiplier variations. It is a sanity check, not part of the proof.

Novelty is provisional. Searches through 13 August 2026 covered the exact question, title, DOI and arXiv id; Ritt/L^1/endpoint/weak-type/counterexample phrases; the local run corpus; the published ETDS version; and nearby Ritt square-function papers. The accepted February 2026 revision still asks the question, and no later answer or matching martingale construction was found. Priority is not claimed. Human review should first check Lemma 1 and the interference bounds (5)–(6).

References

[1] J. Hults and K. Reinhold-Larsson, *Square functions and variational estimates for Ritt operators on L^1*, Ergodic Theory Dynam. Systems (2026), doi:10.1017/etds.2026.10293; arXiv:2507.07256v2.